

QuickHyre AI Associate - What to Learn

Priority: MUST Learn (Class A keywords added to your resume)

- **Machine Learning (ML) fundamentals** — the JD wants general ML literacy even though your background is applied GenAI (RAG, LLM agents), not classical ML. Learn the core vocabulary: supervised vs. unsupervised learning, train/test split, overfitting, and the difference between a model and a pipeline. Andrew Ng’s “Machine Learning Specialization” (Coursera, audit free) intro module covers the 20% that matters for interviews.
- **Deep Learning** — the neural-network side of ML. You already use LLMs (Gemini, OpenAI) as consumers via APIs; this asks for a level below that — understanding roughly how a transformer/neural net is structured (layers, weights, backpropagation at a conceptual level). Fast.ai’s “Practical Deep Learning” first lesson, or 3Blue1Brown’s neural network video series, gets you conversational fast.
- **Neural Networks** — same family as Deep Learning above; know the basic building blocks (neurons, activation functions, layers) well enough to describe how an LLM is built on top of them.
- **Natural Language Processing (NLP)** — you already do NLP-adjacent work (intent handling, RAG semantic search), so this is mostly a labeling exercise: tokenization, embeddings (you already know these from RAG), and classic tasks (classification, NER, sentiment). Bridge: your Convex vector search and Gemini embeddings work IS applied NLP — learn the formal vocabulary to describe it that way.
- **Data Science (as a discipline)** — the JD wants familiarity with working on datasets to support model development. Learn: the basic data-science workflow (explore → clean → feature-engineer → train → evaluate) and pandas/numpy basics for data manipulation. Your SQL/backend-data experience is a head start on the “working with data” half.
- **SQL (preferred)** — you already know this from PostgreSQL work; no gap here, just confirm you can talk about joins, aggregations, and window functions fluently in an interview.

Good to Know (Class B — best-effort, strengthens your fit)

- **Python for ML/data work** — you likely know Python at a basic level (FastAPI/Flask exposure); the delta is the data-science libraries: pandas, numpy, and scikit-learn basics (fit/predict pattern). 3-4 hours with a scikit-learn quickstart covers most interview questions.
- **AI frameworks & LLM familiarity** — you’re already strong here (RAG, LLM tool calling, agentic workflows via Echo/Vibe). Frame this explicitly in interviews as your standout differentiator versus typical fresher candidates.
- **Academic/hackathon AI/ML projects** — if asked, present Echo and Vibe as your equivalent: real, production-grade GenAI projects rather than classroom exercises.

How this maps to your background

This JD is unusually well-aligned with what you’ve already built: RAG pipelines, LLM function/tool calling, prompt engineering, and agentic workflows (Echo, Vibe) are exactly the applied Generative AI skills many “AI Associate” roles want. The gap is on the classical ML/DL/NLP *theory* side — vocabulary and fundamentals you likely haven’t needed for LLM-API-based engineering. Spend your prep time getting fluent in ML/DL/NLP terminology first (highest keyword weight), then a light pandas/scikit-learn pass if time remains — your production GenAI work will do the heavy lifting in the interview itself.